Influence of social factors on economic stability within the EU

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***Abstract*—This study investigates the extent to which social factors - such as tourism, education, healthcare, and quality of life - can predict GDP per capita across European countries. Using data sourced from Eurostat between 2015 and 2023, a clean, harmonised dataset was constructed, and multiple regression and classification models, including Linear Regression, Random Forest, and XGBoost, were employed. Performance was evaluated using RMSE, R^2, precision, recall, and F1-score, with time-series train-test splits to simulate real-world forecasting scenarios.**

Results show that while Linear Regression provided baseline interpretability, Random Forest and XGBoost significantly outperformed it, capturing complex non-linear interactions among social indicators. Feature importance analysis consistently highlighted life expectancy and tourism-related factors as the most influential predictors of GDP per capita. These findings suggest that social sector investments play a critical role in economic resilience and growth, offering valuable insights for policymakers seeking to optimise resource allocation. The study demonstrates the practical utility of machine learning for socio-economic forecasting and underscores the need for a holistic approach to economic policy design.

***Index Terms*—GDP per capita, social indicators, economic forecasting, machine learning, Random Forest, XGBoost, data visualisation, feature importance**

I. Introduction

Economic stability is traditionally assessed through financial metrics such as GDP, trade balances, and industrial output [1]. However, research suggests that social factors such as tourism and education are drivers of long-term economic resilience and can have significant influence [2]. Investments in these areas can contribute to workforce productivity, innovation and consumer spending, yet the accurate impact on economic performance is not well quantified [3]. The majority of current economic models focus mainly on industrial development and financial policies. However, an increasing number of studies support a more holistic view that incorporates social factors into assessments of economic prosperity [4].

This project assesses the extent to which social factors contribute to economic stability by analysing their correlation with GDP per Capita across member EU countries. Through statistical and machine learning modelling, the aim is to evaluate whether stronger social conditions are associated with a strong GDP per capita.

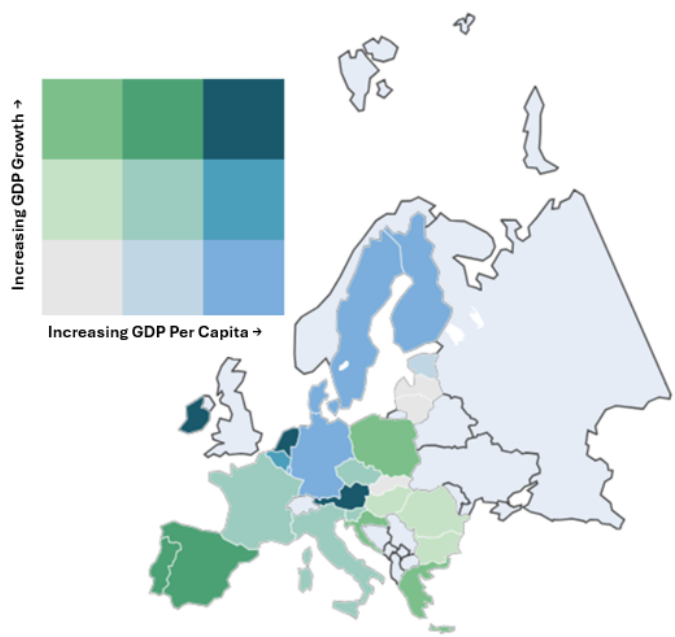

Fig. 1. Distribution of GDP growth and GDP per capita tertiles, 2024

A. Significance of the Study

Figure 1 shows that the EU is still in an economic growth state, giving meaning to the project as the GDP per capita which is being predicted will still increase. This is further supported by the 'European Economic Forecast' in 2023 confirming that the EU's economy is projected to continually grow

[5]. Being able to predict and understand the growth depending on social improvements is crucial, so that policymakers aiming to foster sustainable growth can make the right decisions. If strong correlations exist between social sector improvements and economic resilience, governments may need to prioritise these sectors in future economic planning. This research contributes to economic policy discussions by offering important evidence on how social infrastructure can serve as an economic stabiliser, ensuring that growth is not only driven by financial and industrial factors but also by investments in human capital and well-being.

B. Economic Indicator

Gross Domestic Product (GDP) per Capita was selected as the primary economic indicator for this study. It represents the total economic output of a country divided by its population, providing a standardised measure of an individuals economic contribution. Unlike total GDP, which favours larger economies, GDP per Capita allows for cross country comparisons by adjusting for population size. It is commonly used by institutions such as the World Bank and IMF to classify countries into economic tiers and assess development levels [6].

C. Social Factor Measurements

The role of social factors is examined through four key areas: tourism, education, healthcare, and quality of life. Tourism is assessed based on number of tourism related jobs, the percent of the country contributing towards tourism, average number of nights spent in a country by tourists. Education's impact is measured through employment rate of recent graduates, and the governments total education expenditure. Healthcare is evaluated in terms of the total healthcare expenditure and life expectancy. Finally, quality of life is measured by the percentage of population at risk of poverty.

II. Data Preparation

The datasets for each social factor were gathered through Eurostat's open API [7]. Within each category, multiple datasets were incorporated to capture a diverse range of indicators, ensuring a comprehensive perspective rather than limiting the analysis to a narrow subset of variables. Most datasets were structured in an annual format and contained three key attributes: country name, country code and measured value. "High value datasets" as labelled by Eurostat were prioritised, as specifically these datasets significantly benefit society, the environment, and the economy when used [8]. In line with the focus of this study, the analysis was restricted exclusively to member countries of the EU. Unofficial and former members were removed from the datasets before merging.

A. Data Cleaning and Integration

The datasets were combined using country name and year pairs as the primary key, aligning data points across indicators and the GDP per capita target dataset. To perform this task, a set of data loaders were implemented to fetch and format the data, ensuring all datasets were standardised. The data loaders fetch each dataset as a Pandas data frame through the Eurostat Python client. Each data frame is then filtered to show only annual data, which is stored in a wide year format, before performing a set of variable renaming and pivoting operations. These operations transform each frame to a common long year format that allowed us to merge them seamlessly.

B. Handling Missing Data

The merging operations were performed using outer joins (i.e. country-year pairs that did not appear across all datasets were still included), resulting in a high volume of missing data. The missing values were addressed using linear interpolation, a method that estimates missing points based on fitting a straight line between them and finding the midpoint. This approach was chosen to preserve data continuity while minimising distortions in the overall patterns. Research has shown that linear interpolation outperforms other simple methods such as mean substitution in terms of maintaining data accuracy and reducing estimation errors [9].

However, some countries and datasets had too many missing data points, and so were not included at all. This included countries such as the UK, Turkey, Montenegro and Albania, who recently joined or have left the EU leading to datasets with large gaps.

C. Ensuring Temporal Consistency

To maintain a uniform time range across all variables, only years that were present in all datasets were used. This ensured a consistent time frame for analysis, preventing biases introduced by uneven data coverage. The datasets were collected between 2015 to 2023, as this was the time period with the most overlaps between datasets preventing the use of unnecessary data extrapolation. By implementing these steps, a clean, well-structured dataset was created, allowing for the analysis of the relationship between tourism, education, healthcare, and GDP per capita across EU countries.

III. Data Exploration

To gain an initial understanding of the dataset, a preliminary exploration of the social factors compared with GDP per capita across EU countries was conducted. This involved visualising distributions, identifying patterns, and assessing relationships between variables.

A. Dashboard Creation

Given the wide range of social factors and economic indicators available for analysis, an interactive dashboard was developed using `Plotly.js` [10]. This enables side-by-side visualisation of social factors and economic indicators (GDP growth or GDP per capita) via bivariate choropleth (BC) maps.

BC maps overlay two indicators - in our case an economic indicator mapped along the horizontal axis and a social factor along the vertical axis. Each country is then assigned a colour blend reflecting the level of both the social and economic axis [11].

To optimise the data visualisation, the BC maps were designed following key visual principles, such as using a consistent legend across all maps and a perceptually ordered colour gradient from pale to deep hues to represent increasing values. While most choropleth maps use discrete classification methods, a continuous colour scale was opted for, as gradual, year-by-year trends needed to be visualised, which may be impossible to detect using fixed class boundaries.

The colour key in Figure 2 clarifies how each country's colour should be interpreted. The colours that lie in the region between the two black diagonals represent cases where the economic and social indicators rise together. In this region, the hue shifts from very light pink (where both indicators are low) to deep purple (where both indicators are high). Above the upper guide line the colour drifts toward pure red where the social indicator is high while the economic indicator remains comparatively low and vice versa for below the lower guide line.

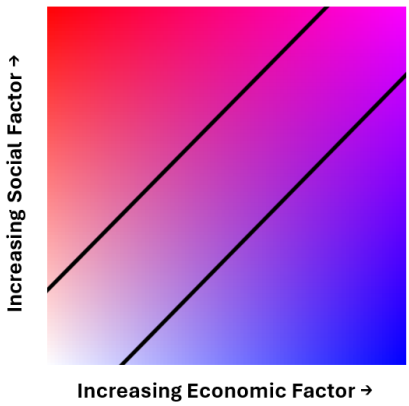

Fig. 2. Colour key for bivariate choropleth maps, where the area between the black lines indicates a proportional relationship between the social and economic factors.

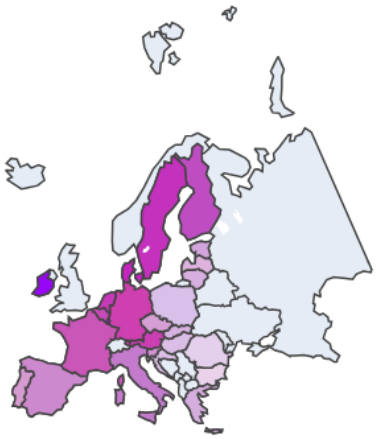

Fig. 3. Relationship between Healthcare Expenditure and GDP per Capita in the EU (2022); lighter colours reflect lower levels of both healthcare expenditure and GDP per Capita.

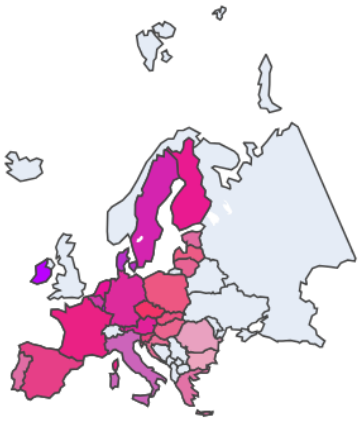

Fig. 4. Relationship between Percentage of Population Engaged in Personal Tourism and GDP per Capita in the EU (2022); deeper colours reflect higher levels of both GDP per Capita and tourism engagement.

By examining these maps, datasets which were incomplete or lacked meaningful correlation were identified. For instance, it became clear that data regarding the number of epidemic cases per year would be insignificant to our analysis. There were so few recorded epidemic cases in the EU over the past decade, that by including such sparse data it could skew or weaken the cumulative dataset.

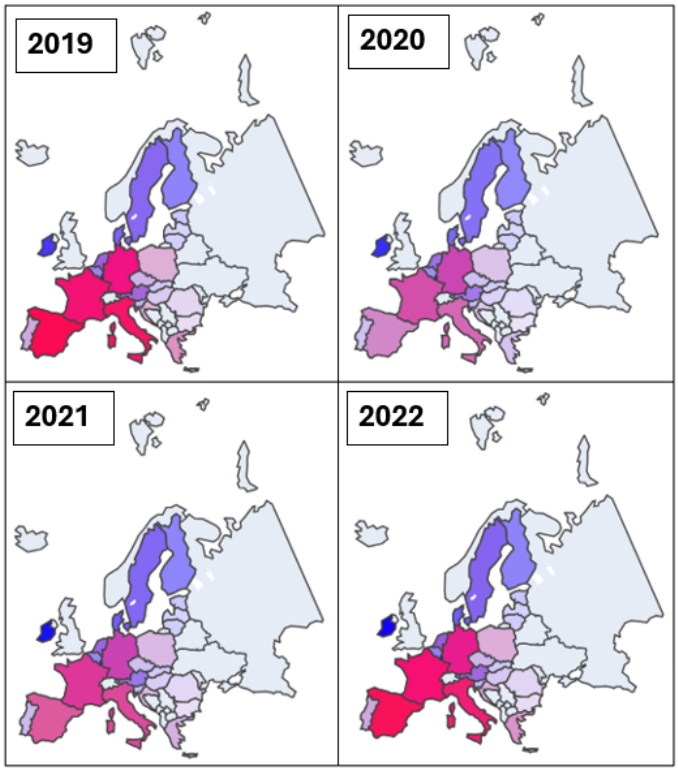

Fig. 5. Nights spent in tourism accommodations across EU countries compared to GDP per Capita (2019–2022); lower red intensity indicates lower GDP per Capita with varying levels of tourism activity, reflecting the impact of COVID-19.

These maps, also enabled us to measure the impact of the

COVID-19 pandemic in 2020: Figure 5 shows the evolution of nights spent at tourist accommodation establishments across the EU from 2019 to 2022. The red hue clearly decreases from 2019 to 2020, reflecting a significant decline in tourism during the COVID-19 pandemic. By 2022, the return of deeper red tones shows that tourism levels had almost recovered to pre-pandemic levels.

Significant other patterns also apparent:

- **Healthcare Expenditure vs. GDP per Capita:** Countries in Figure 3 exclusively display colours ranging from pale pink to deep purple within the central band of the legend. Indicating a strong positive correlation between healthcare expenditure and GDP per capita.
- **Personal Tourism vs. GDP per Capita:** Similarly, Figure 4 highlights that countries with a high percentage of residents engaging in personal tourism also tend to have higher GDP per capita.

The complete interactive dashboard can be accessed at https://europesocialeconomicsdashboard.onrender.com (please allow up to one minute to load). The dashboard allows users to:

1) **Hover Over Countries:** Tooltips display exact numeric values for both GDP metrics and social factors (e.g., tourism percentage, healthcare expenditure).
2) **Filter by Year or Indicator:** Users can change the year or either axis variable to explore how relationships change over time. For instance, comparing pre-pandemic and post-pandemic data illuminates how tourism dipped dramatically in 2020.

Overall, the dashboard visualisation proved valuable for verifying data coverage, informing the direction of our modelling approach and providing an understanding to the relationship between social factors over time.

B. Correlation Analysis

To quantify the correlation of each social factor, three correlation tests measuring the relationship between these social factors and GDP per capita for each country were calculated. An average of each factor's coefficient over each year was taken which gave the final correlation of per social factor to GDP per capita. This was then repeated for each correlation test to explore the robustness of the relationships observed. Using multiple correlation methods ensures that both linear and non-linear associations are captured, and reduces the risk of over reliance on a single statistical assumption. This approach ensures a reliable understanding of how the social factors relate to the GDP per Capita. The coefficients used are:

1) Pearson's Coefficient:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}}, \tag{1}$$

The Pearson correlation coefficient (r) measures the strength and direction of a linear relationship between variables. Values close to either extreme indicate that the social factor changes in a predictable and linear way with GDP. However, this method assumes normally distributed data and is sensitive to outliers [12].

2) Spearman's Coefficient:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}, \tag{2}$$

d_i is the difference of rank between x_i and y_i
n is the number of samples

The Spearman p-value is nonparametric and measures monotonic relationships. By numbering each data point in numerical order, Spearman's correlation uses the difference in rank of the two factors of the data point to output a value.

Importantly, as Spearman finds non-linear but monotonic trends it will be able to detect the relationships of our social factors based on a consistent direction of change without a consistent magnitude [13].

3) Kendall's Coefficient:

$$\tau = \frac{C - D}{\frac{1}{2}n(n-1)}, \tag{3}$$

C is the number of concordant pairs
D is the number of discordant pairs

Like Spearman's coefficient, Kendall's (τ) is a non-linear measure of association. It is based on the concept of concordance between pairs of observations - meaning it compares the ordering of data points between two variables. A pair is considered concordant if the ranks of both variables increase or decrease together, and discordant if one increases while the other decreases. By focusing on the relative order rather than magnitude, Kendall's τ is less sensitive to outliers [14].

4) Results:

For each coefficient, values that are close to -1 or 1 suggest that factor has a high correlation to GDP per capita, either positively or negatively. Whereas factors close to zero suggest little to no correlation.

	Features correlation with 'GDP per Capita':	Pearson	Spearman	Kendall
1	total health care expenditure	0.896377	0.900207	0.821601
2	number of tourism related jobs	0.716693	0.694099	0.638233
3	employment rates of recent graduates	0.520156	0.530323	0.462862
4	life expectancy	0.421129	0.486615	0.441261
5	number of nights spend in country	0.308395	0.433437	0.407039
6	percent of country contributing to tourism	-0.110005	-0.076812	-0.028364
7	government education expenditure as a percent of gdp	-0.223160	-0.202809	-0.194143
8	population percent at risk of poverty	-0.508862	-0.472844	-0.438275

Fig. 6. Comparison of correlation test rankings between social factors and GDP per Capita; strong positive or negative correlations are shown by deeper greens and reds, while lighter colours indicate weak or no correlation.

Across all three correlation measures, a consistent relationship emerged between social factors and GDP per capita. Healthcare expenditure and education indicators demonstrated the strongest positive correlations, reinforcing the hypothesis that investment in human capital brings higher economic performance. Tourism and quality of life measures also showed positive associations, although the strength of these relationships was slightly more variable across countries and years.

Pearsons' correlation, showed that healthcare spending and number of tourism related jobs had the most significant and stable linear associations with GDP per capita. This justified the initial decision to implement a linear regression model for predictive modelling. Meanwhile, Spearmans' and Kendalls' coefficients, confirmed that the observed associations still existed even when the assumptions of only linearity were relaxed. This analysis strengthens confidence in the validity of the results.

Overall, the correlation analysis suggests that stronger social sector development, particularly in health and education, is associated with greater economic prosperity in Europe. These findings support the idea that social improvements can act as important drivers of economic stability and could be used in predictive models.

5) Feature Cross Correlation:

In addition to analysing the correlation between each feature and GDP per capita, it was considered essential to examine the relationships among the features themselves. The rationale behind this was that if two features are highly correlated, they may convey overlapping information, making the inclusion of both potentially redundant [15].

Figure 7 displays the cross-feature correlation matrix for the eight selected variables. Pearson's correlation coefficient, as described previously, was used to quantify the strength of these relationships. It is clear that several features exhibit strong linear correlations. The two strongest relationships are between 'number of nights spent in country' and 'number of tourism-related jobs' (0.86), a naturally expected association, and between 'number of tourism-related jobs' and 'total healthcare expenditure' (0.93), which was less anticipated. This insight allows us to be cautious when developing our machine learning models and to decrease the risk of overfitting.

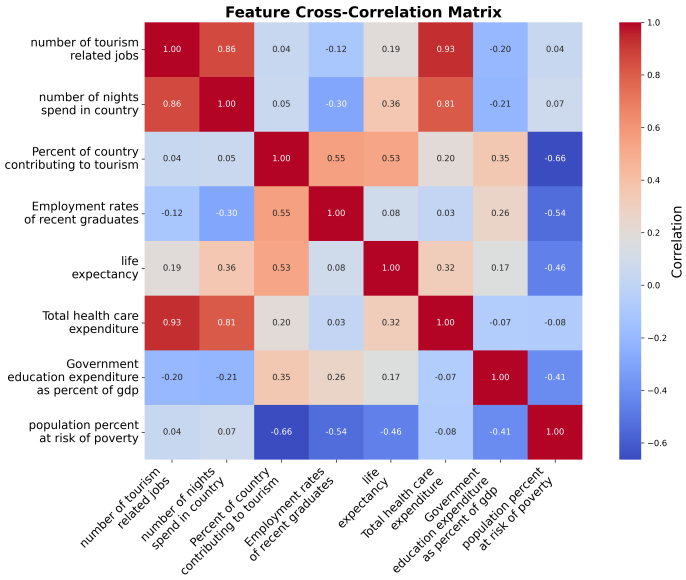

Fig. 7. Cross-correlation of features using Pearson's coefficient; deep reds indicate strong positive linear relationships, while deep blues represent strong negative linear relationships.

IV. Prediction Models

We implemented three models, linear regression, random forest, and Extreme Gradient Boosting (XGBoost) to explore the relationship between social indicators and GDP per capita. Each model was first used to predict a continuous GDP value, allowing for comparison against actual economic data. The same models were then used to classify countries into GDP per capita bands. This classification offers a more interpretable output, enabling broad assessments of national performance without requiring precise economic figures. For example, the World Bank groups countries by GDP per capita into income bands which is a method used in policy-making, lending, and global health funding decisions [16].

A. Linear Regression

From the Pearson correlation analysis it was found that there is a linear relationship between different social factors and GDP per capita. Social indicators, including healthcare spending and education levels, displayed strong linear correlations with GDP per capita. These high scores suggested that a linear model such as linear regression would be suitable. Furthermore, linear regression is a widely used method in economic modelling [17].

Linear regression models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the data. It estimates coefficients that minimise the residual sum of squares between actual and predicted values. The assumption is that the GDP per capita can be expressed as a linear combination of the input features. To address overfitting, Ridge Regression was used, which adds a penalty term to the loss function, improving

stability with more independent variables. The process used scaling to handle outliers and Principal component analysis (PCA) for dimensionality reduction.

B. Random Forest

Building on this, a Random Forest algorithm was implemented. Unlike linear regression, Random Forest does not assume a linear relationship between inputs and the target variable, making it better suited to capture complex and non-linear interactions often found in socioeconomic data [18]. This is shown in the correlation table in Figure 6 where the nonlinear tests gave different results to Pearsons'.

Random Forest is widely used in applied machine learning for its robustness, ability to handle high dimensional data, and resistance to overfitting [19]. It works by constructing an ensemble of decision trees, each trained on a random subset of the data and features, and aggregating their outputs to form a final prediction [20].

The model was implemented using `scikit-learn`, with key hyperparameters such as the number of estimators and tree depth tuned to optimise performance.

C. Extreme Gradient Boosting (XGBoost)

The final model implemented was XGBoost, which builds upon the decision tree structure used in Random Forests. However, unlike Random Forests that construct trees in parallel on random subsets of data and features, XGBoost builds trees sequentially, where each new tree attempts to correct the errors of the previous one. This boosting approach allows the model to achieve higher accuracy through focused learning on difficult cases. XGBoost is a scalable and regularised gradient boosting algorithm introduced by Chen and Guestrin [21]. It is known for its speed, flexibility, and superior performance on structured data, and has become a popular choice in applied machine learning tasks.

Key hyperparameters tuned included tree depth, learning rate, number of estimators, and regularisation parameters to ensure a strong model performance.

V. Model Performances

Before developing the predictive models, it was essential to clearly define both the modelling objective and the evaluation criteria. The goal was to predict GDP value in the year following the training period for a given country.

To assess model performance in regression, two standard regression metrics were used: Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). These metrics were selected based on their widespread use in similar research contexts [22]. RMSE provides a measure of the average magnitude of prediction errors, while R^2 quantifies the proportion of variance in the target variable that is explained by the model.

For the classification task, the model predicts across four classes, as shown in Table I. The classes were defined using quantile-based binning, which splits the range of GDP per capita values into four equally populated quartiles based on the training data. This approach ensures that each class contains a similar number of countries, helping to maintain balance during classification.

TABLE I
GDP per Capita Class Bands and Corresponding Class Labels

Class Label	Lower Bound (€)	Upper Bound (€)
0	8,130	17,109
1	17,110	26,619
2	26,620	45,509
3	45,510	107,849

Precision, recall, and F1-scores were used to evaluate each classification model's performance. The F1-score, in particular, is well-suited for imbalanced class distributions, as it balances the trade-off between precision (the proportion of correct positive predictions) and recall (the proportion of actual positives correctly identified). This makes it a more informative metric than accuracy alone, especially when misclassifications in minority classes carry greater importance [23].

Each model was trained using a time series split approach, where training occurred on historical data and testing was conducted on subsequent periods . This method better reflects real-world forecasting scenarios, where future data is unknown at training time [24]. Specifically, models were trained on the time range 2015–2022 and were tested on data from 2023. This results in an approximate 89%–11% train-test split, which leans slightly towards a larger training proportion than typical practice for datasets of moderate size [25]; however, it remains appropriate given the temporal structure of the data and the need to simulate real-world forecasting conditions.

After initial training, each model's performance was further refined through hyperparameter tuning. This was accomplished using a grid search strategy, which evaluates a predefined set of parameter combinations to determine the configuration that yielded the best results based on validation performance.

The objective of these models extends beyond achieving strong predictive performance; they also aim to provide insights into the key real-world drivers of GDP per capita. By analysing the feature importances of the best-performing models, it becomes possible to infer which factors exert the greatest influence on economic outcomes.

A. Regression Model Results

The results of the predictive modelling are summarised in Table II, which compares the performance of the three models on 2023 GDP per capita estimates using RMSE and R^2.

TABLE II
Comparison of Model Performance on GDP per Capita

Metric	Linear Regression	Random Forest	XGBoost
RMSE	14001.66	7343.92	4866.06
R^2	0.6304	0.9071	0.9554

As shown in Table II, the XGBoost model achieved the best predictive performance, with the lowest RMSE and the highest R^2 value. The high R^2 scores of the Random Forest (0.91) and XGBoost (0.96) models indicate that they explain a substantial proportion of the variance in GDP per capita across countries. These results suggest a high degree of predictive accuracy, with relatively low average error compared to the scale of the target variable [26].

To visualise and assess the performance of each model, Figure 8 presents a scatter plot comparing the predicted and actual GDP per capita values for 2023 across all countries in the dataset.

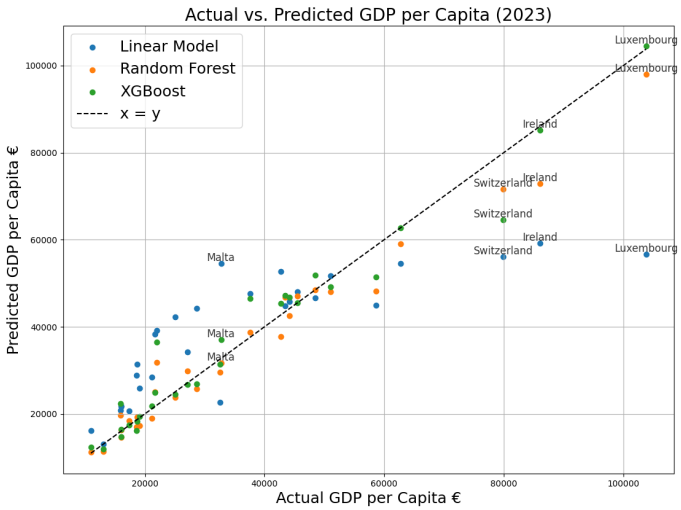

Fig. 8. Scatter plot comparing actual vs predicted GDP per capita for all three models. The diagonal line ($x = y$) indicates perfect prediction, where predicted values equal the actual values.

In Figure 8, data points that lie close to the diagonal line ($x = y$) indicate strong predictive accuracy, as the predicted values closely match the actual values. The XGBoost and Random Forest models exhibit tight clustering around this line, suggesting well-calibrated predictions with minimal bias.

By contrast, the Linear Regression model displays a noticeably looser fit, with a clear tendency to under-predict higher GDP per capita values, particularly for countries such as Switzerland, Ireland, and Luxembourg. Notably, it failed to predict any values exceeding €60,000, highlighting a limitation in its ability to capture non-linear relationships present at the upper end of the income distribution. Another standout observation was the model's poor performance in predicting Malta's GDP per capita, which may be due to the country's unique economic structure or outlier characteristics that are not well captured by a simple linear trend

The superior performance of XGBoost and Random Forest in handling higher-value predictions suggests that these models are better suited to capturing the potential more complex patterns within the data.

To further assess model performance, Kernel Density Estimates (KDEs) of actual and predicted GDP per capita values are shown in Figure 9. KDE is a non-parametric method for estimating the probability density function of a continuous variable, producing smooth, continuous curves that reveal the shape of the distribution. Unlike histograms, KDEs avoid binning and extend slightly beyond the observed data range due to the smoothing process [27].

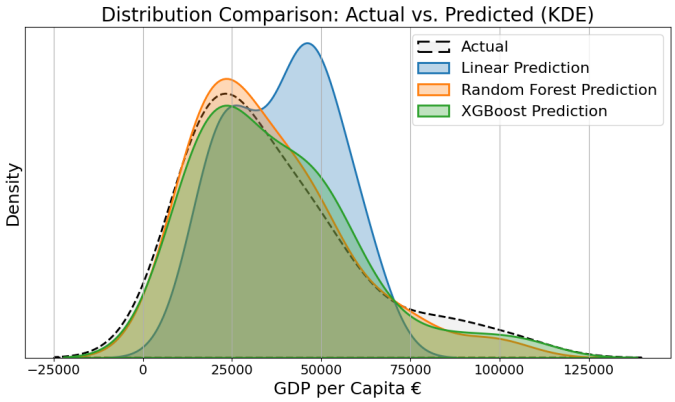

Fig. 9. Kernel Density Estimate (KDE) plot of predicted GDP per capita values for all three models, with the actual distribution shown as a black dotted line.

In a regression setting, Figure 9 is particularly useful for comparing the predicted distribution to the actual one, helping to identify potential biases, skewness, or over-regularisation [28]. In our case, the predicted KDEs for the Random Forest and XGBoost models closely align with the actual distribution, indicating that these models not only fit individual values accurately but also approximate the overall structure of the data effectively. By contrast, the Linear Regression model again underperforms, with a predicted distribution that diverges more noticeably from the actual values.

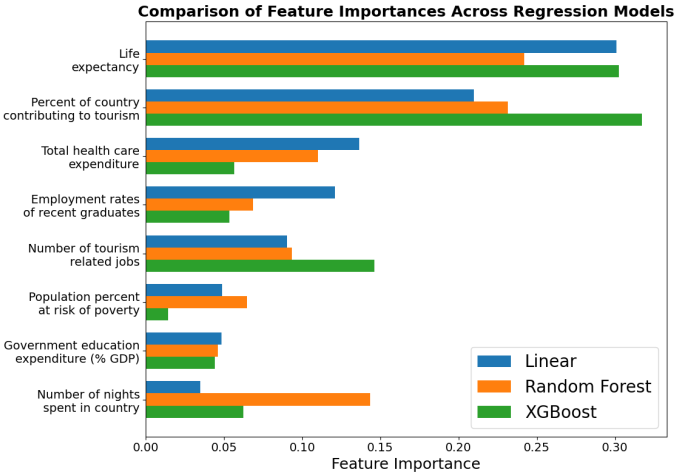

Fig. 10. Feature importance ranking for the regression models, indicating the relative contribution of each social factor to model predictions. Higher bars denote greater influence on GDP per capita estimation.

The feature importance analysis in Figure 10 shows that the percentage of a country contributing to tourism and life

expectancy were the most influential predictors of GDP per capita. This aligns with expectations, as economic development is often closely linked to health and service-related indicators [29].

Interestingly, all three models consistently identified these variables as the most significant, while also agreeing that indicators such as the population at risk of poverty and government expenditure on education contributed less to predictive performance.

Taken together, the results confirm that the XGBoost model is a slightly more accurate and better-calibrated extension of the Random Forest model, producing outputs that closely reflect the real-world distribution of GDP per capita. The Linear Regression model, however, struggles to compete - likely due to its limited capacity to capture complex, non-linear relationships within the data.

B. *Classification Model Results*

The results of the classification predictive models are summarised in Table III, which compares the performance of the three models on 2023 GDP per capita estimates using macro-averages and overall accuracy.

TABLE III
COMPARISON OF MACRO-AVERAGED CLASSIFICATION METRICS AS WELL AS OVERALL ACCURACY BETWEEN MODELS

Metric	Linear Regression	Random Forest	XGBoost
Precision	0.694	0.837	0.837
Recall	0.609	0.854	0.854
F1-Score	0.616	0.827	0.827
Accuracy	0.621	0.828	0.828

As shown in Table III, both Random Forest and XGBoost performed considerably better than the Linear Regression model across all the macro-averaged measures, which calculate the mean performance equally across all classes, regardless of class imbalance. The tree-based algorithms had macro-averaged precision, recall, and F1-scores of 0.837, 0.854, and 0.827, respectively, as well as a generalised classification accuracy of 0.828. On the other hand, the adapted Linear Regression model, which rounds its continuous results for classification, had lower scores for all measures.

Most notably, the Random Forest and XGBoost models yielded the same results, highlighting the robustness of ensemble tree-based approaches to identifying the inherent structure within the feature space. Lower performance of the Linear model stems once again from its poor ability to model non-linear relationships, which have consistently proven to be of the utmost importance for classifying countries into GDP per capita quartiles.

TABLE IV
CONFUSION MATRICES ACROSS MODELS (COLUMNS REPRESENT PREDICTED CLASSES 0–3, ROWS REPRESENT ACTUAL CLASSES)

Linear Regression

$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 5 & 3 & 0 \\ 0 & 1 & 5 & 3 \\ 0 & 0 & 1 & 6 \end{bmatrix}$$

Random Forest

$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 2 & 6 & 0 & 0 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

XGBoost

$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 2 & 6 & 0 & 0 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

The corresponding confusion matrices provided in Table IV back up these results. For Linear Regression, there is heavy misclassification of the elements of Class 1 and Class 2 with their predictions spread far afield to their next nearest categories. Random Forest as well as XGBoost models, however, were almost perfect for classifying Class 0 as well as Class 3. Random Forest as well as XGBoost models having the same confusion matrices reveal their strength at learning the class structure of the data as well as reducing inter-category errors.

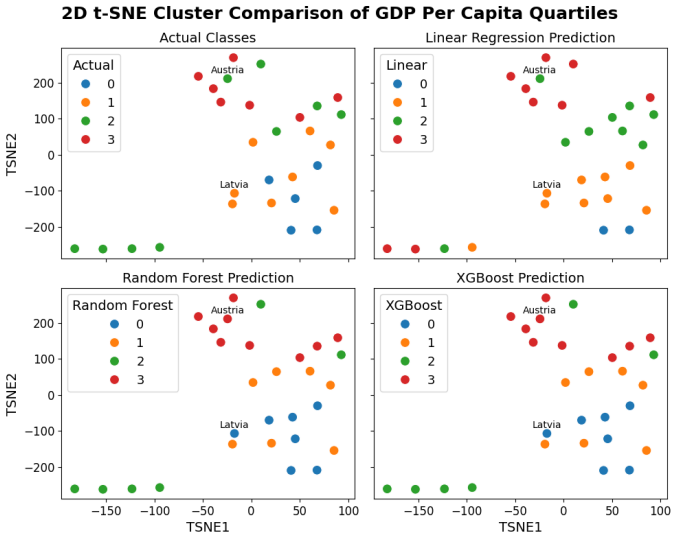

Fig. 11. Two-dimensional t-SNE visualisation of the test data, showing how countries cluster based on social indicators.

In order to visualise how these classes were separated and to further analyse the confusion matrix results, t-distributed Stochastic Neighbour Embedding (t-SNE) was employed. t-SNE is particularly well-suited for projecting high-dimensional data into a two-dimensional space while preserving local structures and neighbourhood relations within the data [30]. The Linear Regression model has a low cohesion among its predictions and therefore more overlapping class clusters. The Random Forest and XGBoost models have more distinct and well-separated clusters, which indicate their ability to identify the complex underlying feature space structure better. The sophisticated models were not able to well predict the class of some observations, however, e.g., Austria and Latvia. This persistent misclassification shows that some cases remain challenging due to similar characteristics between GDP per capita quartiles.

Following the cluster-based analysis provided by the t-SNE visualisation, a deeper examination of feature importances was conducted to understand the underlying drivers behind the classification outcomes. Figure 12 displays the comparison of feature importances across the Linear Regression, Random Forest, and XGBoost models. It is evident that 'Life expectancy' and 'Percent of country contributing to tourism' consistently emerge as the most influential features across all three models, with life expectancy being particularly dominant.

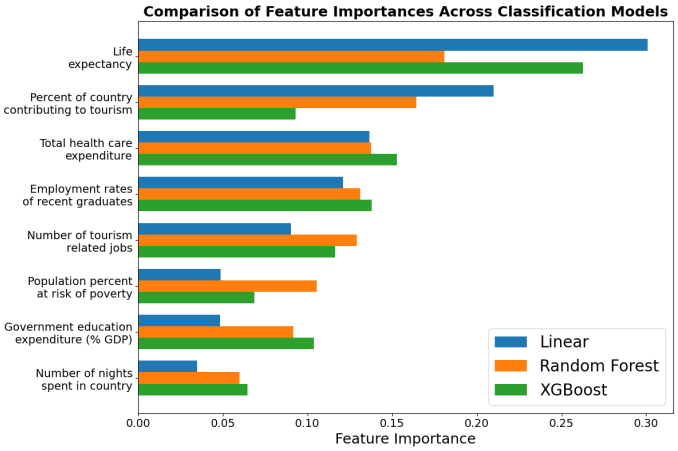

Fig. 12. Feature importance ranking for the classification models, indicating the relative contribution of each social factor to model predictions. Higher bars denote greater influence on GDP per capita estimation.

The patterns of feature importance offer essential background for understanding the misclassifications experienced in the improved Random Forest and XGBoost models. While these ensemble approaches reported higher overall accuracy, taking a closer look at individual misclassified instances shows how small differences in feature values can cause mistakes.

For example, Austria was misclassified as Class 3 instead of its true Class 2. Its 'Percent of country contributing to tourism' value of 76.66 was notably closer to the Class 3 training average of 79.12 than to the Class 2 average of 65.07, making it more challenging for the model to assign correctly. Similarly, Latvia was incorrectly classified into Class 1 rather than Class 2, partly due to its 'Life expectancy' of 75.6, which aligned more closely with the Class 1 average of 76.34 than the Class 2 average of 80.06.

These examples show how even strong models can struggle with border-line cases in which individual feature values deviate from typical class patterns. Thus, the feature importance analysis not only explains the overall performance differences observed among models but also explains the specific challenges faced when classifying countries with atypical profiles.

VI. Discussion

A. Comparison of Models

Across both regression and classification tasks, XGBoost consistently outperformed Random Forest and Linear Regression. In regression, XGBoost achieved the lowest RMSE (4866.06) and highest R^2 (0.9554), indicating strong ability to model complex relationships between social factors and GDP per capita. Random Forest followed with a respectable R^2 of 0.9071, while Linear Regression showed clear limitations (R^2 = 0.6304), likely due to its inability to capture non-linear patterns. For classification, both XGBoost and Random Forest achieved high F1-scores (0.827) and accuracies (0.828), significantly outperforming Linear Regression (F1 = 0.616, accuracy = 0.621). t-SNE visualisations confirmed that tree-based models more accurately reproduced the structure of the true GDP class labels, while Linear Regression showed greater class overlap.

Feature importance analyses were consistent across models, with life expectancy and tourism-related indicators emerging as key predictors. The tree-based models further highlighted the complex interactions between features that linear models could not capture.

In summary, while Linear Regression provided useful baseline interpretability, Random Forest and particularly XGBoost delivered superior predictive performance, better class separation, and more reliable identification of influential factors.

B. Model Limitations

Although the models achieved strong predictive performance, several limitations remain. Linear regression assumes linearity and normally distributed residuals, assumptions which do not fully hold in the real-world. As a result, some non-linear patterns are missed.

Complex models like XGBoost capture non-linear relationships but increase the risk of overfitting and require careful hyperparameter tuning. Both approaches assume stability in the underlying data distribution, making them sensitive to major economic shifts or shocks such as COVID-19.

Finally, there is a trade-off between model complexity and interpretability. Linear regression offers transparent coefficients linked to social factors, while XGBoost, even though it has greater accuracy, produces less interpretable outputs. These factors must be considered before deploying models in the real-world where one would be presenting to officials/executives who have limited knowledge on the subject.

C. Data Challenges and Solutions

The primary challenge was the inconsistency and incompleteness of social indicators across countries from 2015 to 2023. Variability in national data collection and occasional gaps in individual features significantly reduced the usable sample size and constrained the range of indicators that could be included in the analysis. As a result, many potentially informative datasets had to be excluded entirely. These issues were mitigated through a robust preprocessing pipeline, which included median imputation and linear interpolation to address missing values, and PCA to reduce feature redundancy while retaining core variance across countries.

D. Real-World Relevance

Beyond continuous predictions, the classification of countries into GDP bands adds practical interpretability. This

mirrors real-world applications such as the World Bank's income groupings, which influence development funding, trade negotiations, and policy benchmarking. Grouped outputs are easier to communicate to policymakers and stakeholders who may not require precise economic estimates. This dual-purpose modelling structure enhances both strategic usability and impact-driven decision-making.

E. Future Work

This modelling framework could support early-stage economic forecasting in countries with limited access to traditional fiscal data. It also has potential for integration into decision-support tools for governments, NGOs, and development agencies, enabling near real-time monitoring of socio-economic trends. Future development could include a web-based dashboard for interactive visualisation, allowing stakeholders to explore predictions and classifications without technical knowledge. Additionally, applying the models to COVID-era data could help test resilience under economic shock conditions, highlighting the sensitivity and responsiveness of social indicators in predicting downturns.

VII. Conclusion

This project has demonstrated that key social indicators are strongly associated with GDP per capita across EU countries and can be effectively applied to forecast economic performance and stability. Comprehensive statistical analysis confirmed that social factors such as education, healthcare, tourism, and quality of life exhibit strong correlations with GDP per capita, indicating that countries with stronger social foundations tend to achieve greater economic prosperity. Leveraging these relationships, predictive models were trained on 2015 to 2022 data and achieved a high degree of accuracy in estimating GDP per capita for 2023, with complex models outperforming the linear baseline model by capturing non-linear patterns in the data.

The introduction of a classification scheme that grouped countries into GDP per capita bands further enhanced the interpretability of the results, allowing quantitative predictions to be translated into accessible categories. Overall, the findings underscore the value of integrating social indicators with traditional economic metrics. This holistic approach yielded more robust and insightful models of GDP per capita, highlighting that social development and economic prosperity are deeply interconnected. The success in predicting 2023 GDP outcomes validates the practical utility of this integrated approach and reinforces its relevance for policymakers and analysts seeking a more comprehensive understanding of economic stability.

References

[1] M. Cangiano, T. Ter-Minassian, and J. Curristine, "Public Financial Management and Its Emerging Architecture", International Monetary Fund, 2013.

[2] B. Wang and T. Chen, "Social Progress beyond GDP: A Principal Component Analysis (PCA) of GDP and Twelve Alternative Indicators", Sustainability, vol. 14, no. 11, pp. 6430, 2022.

[3] K. Arrow, "The economic implications of learning by doing," Review of Economic Studies, vol. 29, no. 3, pp. 155–173, 1962.

[4] J. Stiglitz, A. Sen, and J.-P. Fitoussi, "Report by the Commission on the Measurement of Economic Performance and Social Progress", 2009.

[5] European Commission, "European Economic Forecast: Autumn 2023", Institutional Paper 267, Directorate-General for Economic and Financial Affairs, 2023.

[6] World Bank, "World Development Indicators 2023", World Bank Group, 2023.

[7] "Eurostat." Accessed: Apr. 26, 2025. [Online]. Available: https://ec.europa.eu/eurostat/web/main/home.

[8] European Commission, "Implementing Regulation (EU) 2023/138: High-Value Datasets under the Open Data Directive", Official Journal of the European Union, 2023.

[9] M. M. Rahman and M. N. Hasan, "Comparison of Linear Interpolation Method and Mean Method to Replace the Missing Values in Environmental Data Set, ResearchGate, 2014.

[10] Plotly Technologies Inc., "Plotly.js: Open-Source JavaScript Graphing Library," [Online]. Available: https://plotly.com/javascript/

[11] C. A. Brewer and L. W. Pickle, "Evaluation of Methods for Classifying Epidemiological Data on Choropleth Maps in Series," Annals of the Association of American Geographers, vol. 92, no. 4, pp. 662–681, 2002.

[12] D. J. Sheskin, "Handbook of Parametric and Nonparametric Statistical Procedures," 5th ed., CRC Press, 2011.

[13] J. H. Zar, "Spearman Rank Correlation", Encyclopedia of Biostatistics, pp. 1–5, 2005.

[14] M. Gibbons and P. E. Chakraborti, "Nonparametric Statistical Inference," 5th ed., CRC Press, 2010.

[15] I. Guyon and A. Elisseeff, "An Introduction to Variable and Feature Selection," Journal of Machine Learning Research, vol. 3, pp. 1157–1182, 2003.

[16] World Bank, "World Development Indicators," The World Bank Group, 2023. [Online].

[17] D. Gujarati and D. Porter, *Basic Econometrics*, 5th ed., McGraw-Hill Education, 2009.

[18] L. Breiman, "Random Forests", *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[19] G. Biau, "Analysis of a Random Forests Model", *Journal of Machine Learning Research*, vol. 13, pp. 1063–1095, 2012.

[20] L. Pal, "Random Forest Classifiers", *eScholarship University of California*, 2005.

[21] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[22] R. Espinoza, F. Fornari, and M. J. Lombardi, "The role of financial variables in predicting economic activity," Journal of Forecasting, vol. 31, no. 1, pp. 15–46, 2012.

[23] J. Davis and M. Goadrich, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," Proceedings of the 23rd International Conference on Machine Learning (ICML), pp. 233–240, 2006.

[24] M. Hasanov, M. Wolter, and E. Glende, "Time series data splitting for short-term load forecasting," PESS+ PELSS 2022; Power and Energy Student Summit, pp. 1–6, 2022.

[25] V. R. Joseph, "Optimal ratio for data splitting," Statistical Analysis and Data Mining: The ASA Data Science Journal, vol. 15, no. 4, pp. 531–538, 2022.

[26] D. Chicco, M. J. Warrens, and G. Jurman, "The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation," PeerJ Computer Science, vol. 7, pp. e623, 2021.

[27] S. Weglarczyk, 'Kernel density estimation and its application,' ITM Web of Conferences, vol. 23, pp. 00037, 2018.

[28] D. Ruppert and D. B. H. Cline, "Bias reduction in kernel density estimation by smoothed empirical transformations," The Annals of Statistics, pp. 185–210, 1994.

[29] A. Case, "Health, income and economic development," Proceedings of the World Bank Conference on Development Economics, 2001/2002, pp. 221–241, 2002.

[30] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," Journal of Machine Learning Research, vol. 9, no. 11, 2008.

VIII. Personal Contribution

I attended all group meetings both mandatory and set up by us. Contributed and lead group discussion with ideas and brainstorming of our project. Completed introduction, model write up section, discussion and conclusion. Also, did a full read through improving all other sections. I created some of the main datasets manually at the beginning to help with first-stage implementations of the ML models.

IX. Strengths and Weaknesses of Project

A. Strengths

- **Data preparation** Used Python data-loaders to automate Eurostat downloads, clean, reshape and merged them into a complete dataset ready to be used.
- **Dashboard** A Plotly-Dash web application which made exploration of the data set much easy by letting users have compared different trends with the datasets.
- **Correlation analysis** Used multiple correlation tests intially to see if there is any correlation between the social factors and GDP per Capita, before trying to build ML models using the data.
- **Strong predictive models** Tested multiple ML models (linear, Random Forest, and XGBoost), showed the non-linear models worked best. XGBoost acheived a $R^2 \approx 0.96$ (regression) and macro-F1 ≈ 0.83 (classification); these accurate results allowed us to generate credible GDP per Capita predictions.

B. Weaknesses / Areas for Improvement

- **Data gaps** - Use of linear interpolation (even though not much) may have smoothed away important data changes, especially for smaller economies. More advanced imputation techniques could have improved this.
- **Limited social features** - We had to exclude some datasets due to missing values.
- **Model interpretability** - XGBoost model gave the best accuracy, but the outputs were less interpretable.
- **Narrow time window (2015–2023)** - Training and testing on less than a decade of data limits the accuracy of models, however there were barely any datasets going back that far, so maybe the use of other websites to collect data could have helped.

X. How the Project Relates to Ideas Taught on the Unit

In our project, we demonstrate how we have met the following learning outcomes:

- **"To acquire a working knowledge of practical data science, applied to real world problems."** We implemented a full data science workflow. Retrieved raw data from the Eurostat, completed data prep, engineering and exploration and then trained and evaluated models to predict a real-world economic indicator.
- **"To be able to start from raw data and deliver a representation that allows a better understanding of the topics in the data."** Cleaned, interpolated, and merged multiple raw data sources into a single long-format dataframe which was then explored through the use of a dashboard.
- **"To have experience of using software tools for data pre-processing and management."** Used Python and Pandas for all preprocessing and imputation.
- **"To acquire first-hand experience in specific techniques for data storage."** Stored processed data in CSVs and Pandas DataFrames.
- **"To understand the differences between different visualisation strategies to efficiently explore the data."** Used scatter plots, KDEs, cross-correlation heatmaps, and bivariate heatmaps to explore the data thoroughly.
- **"To have learned how to present and interpret data to/for a non-technical audience."** Created an interactive dashboard with maps, time-series graphs, and clear legends.
- **"To be able to share data under privacy constraints."** We used only country-level statistics and avoided any individual-level datasets.
- **"To have practiced teamwork and time management."** We had weekly group meetings, divided tasks equally and complete work on dealines set, while collaborating on the planning, preparation/exploration, modelling, presentation and documentation.

XI. Achievements and What the Project Has Taught Me

- **Machine Learning Modelling** - Researched and worked with multiple ML techniques (including classification, regression, clustering, ensemble and dimensionality reduction methods), and applied t-SNE clustering, linear & ridge regression, Random Forest, and XGBoost to predict GDP per capita, which improved my understanding of model selection and hyperparameter tuning.
- **Web Scraping** - Learned about Beautiful Soup and REST APIs, although the Eurostat data turned out to be readily available from direct API calls or by downloading the dataset.
- **Data Ethics and Privacy** - Learned about ethical considerations and privacy constraints when working with real-world data. This taught me to limit our analysis to aggregated, country-level indicators and to document all imputation steps transparently to maintain data integrity.
- **Data Management and Wrangling** - Learned and practiced Pandas operations for cleaning, merging, and reshaping data to create a single, analysis-ready dataset.
- **Data Exploration** - Learned about how to use PCA for dimensionality reduction, and correlation analysis (Pearson, Spearman, Kendall) to find relationships in the merged dataset.
- **Data Visualisation** - Learned the basics of Matplotlib and Seaborn, and learned Plotly and Plotly.Dash in depth to create the interactive dashboard.